Friedrich-Alexander-Universität
Technische Fakultät

Project Part 1

Computational Visual Perception (CompVP)

Bernhard Egger, Andreas Kist, Patrick Krauß, Tim Weyrich

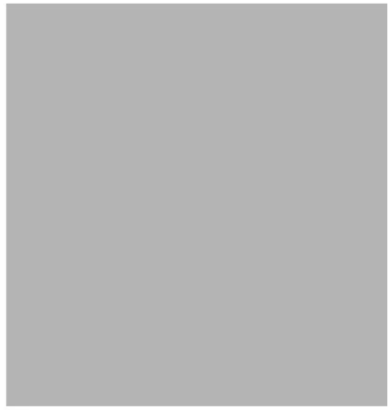


3D Reconstruction: Analysis-by-Synthesis

Input Image



Synthetic Image



$$\leftrightarrow \quad = \mathrm{R}_{\substack{pose,\\ illumination}} \left(\alpha_1 \cdot \; + \alpha_2 \cdot \; + \alpha_3 \cdot \; + \ldots \right)$$

$$\left(\beta_1 \cdot \; + \beta_2 \cdot \; + \beta_3 \cdot \; + \ldots \right)$$

Minimize image difference by iterative non-linear optimization

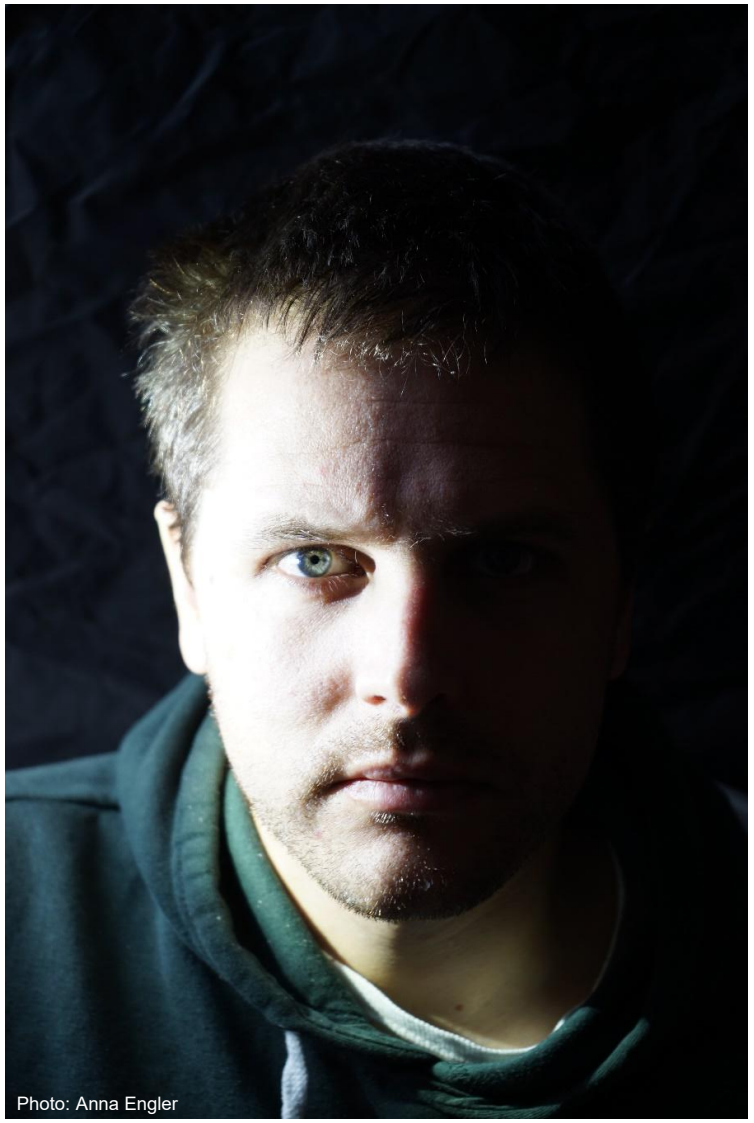
Output:
3D model



Scene Understanding

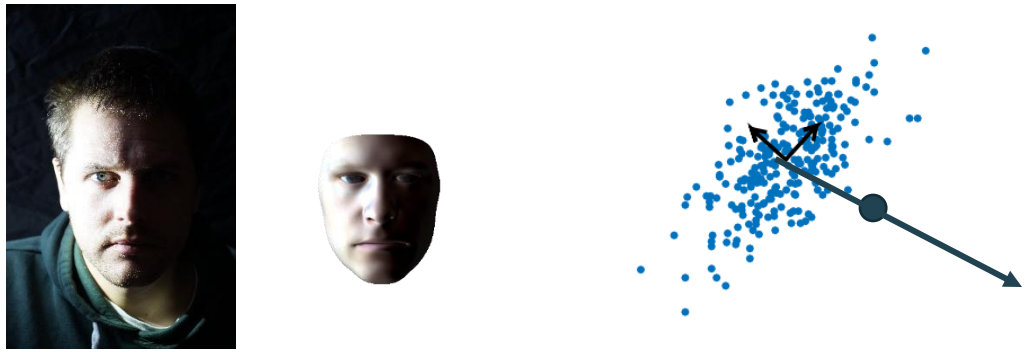


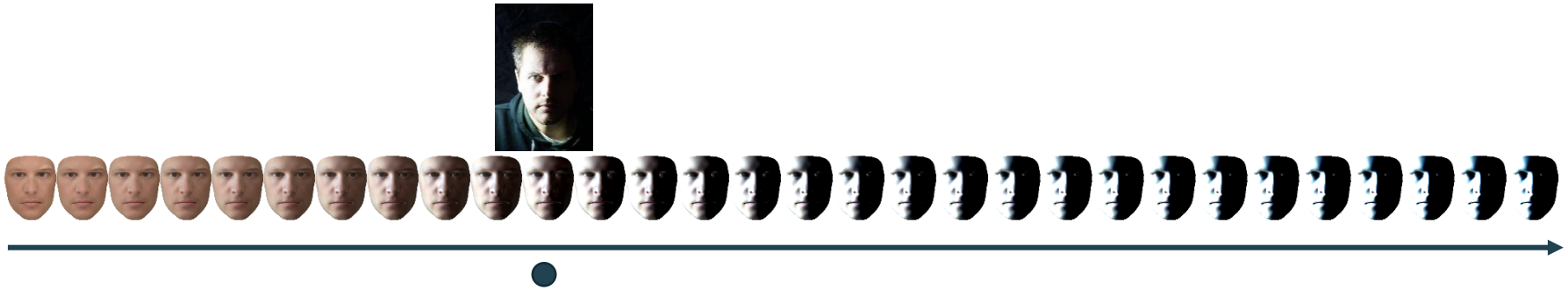
Generalization, but how?



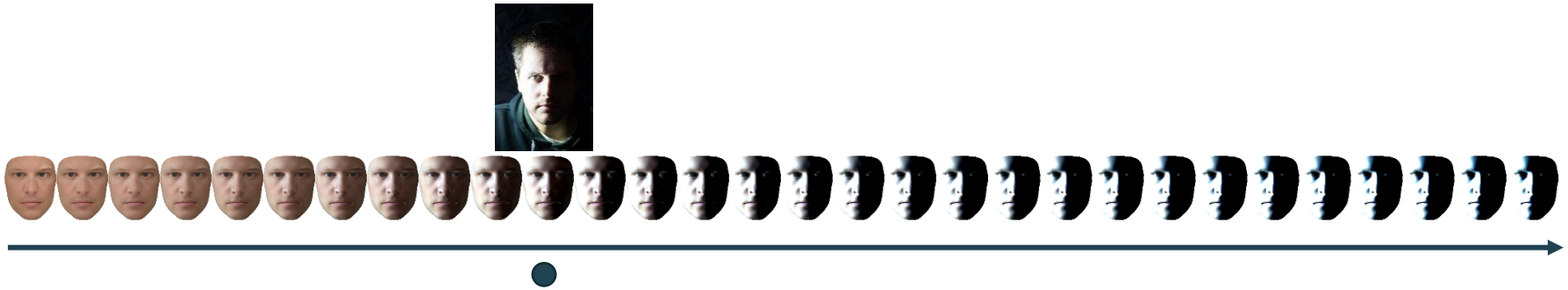


Illumination extrapolation





Mooney reconstruction?









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What if we change contrast polarity?













How to complete module

- 7,5 ECTS
  - 5+2,5 ECTS
  - You need both!
  - You can’t get only 5 or only 2,5 ECTS

Overall project goal

- How well do state of the art vision systems understand Mooney images?
- How do biologically constrained vision systems perceive Mooney images?

Paper 1 by October 27th :

Inverse Rendering Best Explains Face Perception Under Extreme Illuminations

Bernhard Egger[1] **Max Siegel**[1] **Riya Arora**[1] **Amir Arsalan Soltani**[1] **Ilker Yildirim**[2] **Josh Tenenbaum**[1]
[1]Massachusetts Institute of Technology [2]Yale University

Abstract

Humans can successfully interpret images even when they have been distorted by significant image transformations. Such images could aid in differentiating proposed computational architectures for perception because while all proposals predict similar results for typical stimuli (good performance), they differ when confronting atypical stimuli. Here we study two classes of degraded stimuli – Mooney faces and silhouettes of faces – as well as typical faces, in humans and several computational models, with the goal of identifying divergent predictions among the models, evaluating against human judgments, and ultimately informing models of human perception. We find that our top-down inverse rendering model better matches human percepts than either an invariance-based account implemented in a deep neural network, or a neural network trained to perform approximate inverse rendering in a feedforward circuit.

Keywords: Mooney Faces, 3D Morphable Model, Inverse Rendering, Analysis-by-Synthesis, Face Recognition, Generalization

Introduction



Figure 1: Despite extreme loss of image information, we can still perceive rich three-dimensional shape and identity for Mooney faces and other objects under unusual illumination. 3D data: MPI SMAL (Zuffi et al., 2017), BFM (Gerig et al., 2018) and MPI FAUST (Bogo et al., 2014)

(Yamins et al., 2014; LeCun, Bengio, & Hinton, 2015). Deep convolutional neural networks (DCNNs) provide a powerful substrate for implementing this approach in practice. Its primary drawback is that learning invariant representations requires large amounts of relevant data, representative of all the problems the system will ever solve. For image distortions not present in the training data, there is no reason to expect such an approach to generalize successfully.

An alternative approach to perception is based on modeling the causal process that generates (or renders) images from the underlying scene, including ways that process could take on extreme characteristics not typically encountered during any training. Conceptually, if we know how to generate an image then we should be able to recognize it by inverting the rendering process – up to noise/statistical constraints and computational difficulties, which may be substantial. In such settings, no additional training is required.

Here we present experiments and concrete model implementations to test these two accounts of generalization in human face perception against each other. To evaluate the inverse rendering (IR) account, we consider several models based on top-down approximately Bayesian (Markov Chain Monte Carlo; MCMC) inference in a well-known generative face model, as well as a recently proposed feedforward network approximation to Bayesian inverse graphics (the EIG model (Yildirim, Belledonne, Freiwald, & Tenenbaum, 2020)). We contrast these IR models with an invariance account based on DCNNs trained for face recognition.



Paper 2 by November 24th :

Article

Connectome-constrained networks predict neural activity across the fly visual system

https://doi.org/10.1038/s41586-024-07939-3

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Open access

Check for updates

Janne K. Lappalainen[1,2,3], Fabian D. Tschopp[3], Sridhama Prakhya[3], Mason McGill[3,4], Aljoscha Nern[3], Kazunori Shinomiya[3], Shin-ya Takemura[3], Eyal Gruntman[3,5], Jakob H. Macke[1,2,6] & Srinivas C. Turaga[3]

We can now measure the connectivity of every neuron in a neural circuit[1–9], but we cannot measure other biological details, including the dynamical characteristics of each neuron. The degree to which measurements of connectivity alone can inform the understanding of neural computation is an open question[10]. Here we show that with experimental measurements of only the connectivity of a biological neural network, we can predict the neural activity underlying a specified neural computation. We constructed a model neural network with the experimentally determined connectivity for 64 cell types in the motion pathways of the fruit fly optic lobe[1–5] but with unknown parameters for the single-neuron and single-synapse properties. We then optimized the values of these unknown parameters using techniques from deep learning[11], to allow the model network to detect visual motion[12]. Our mechanistic model makes detailed, experimentally testable predictions for each neuron in the connectome. We found that model predictions agreed with experimental measurements of neural activity across 26 studies. Our work demonstrates a strategy for generating detailed hypotheses about the mechanisms of neural circuit function from connectivity measurements. We show that this strategy is more likely to be successful when neurons are sparsely connected—a universally observed feature of biological neural networks across species and brain regions.

Structure of project

- Project Part 1: Play around with Mooney faces and computer vision models
  (close to Paper 1)

- Project Part 2: Play around with a simulated visual system of a fly (close to paper 2)

- Project Part 3: Further Evaluation based on results of Part 1 and Part 2
  (close to paper 1 and 2)

How to pass

- Submission via Studon course:
  “Computational Visual Perception”

- Submission has to be in the exact format

- Three strict project deadlines
  - November 20st
  - January 8th (feel free to submit early)
  - February 5th

How to pass

- Pass/Fail for each of the 3 parts,
- You need to pass 2 out of 3 parts

- The best solutions for each part will be released ~ 1 week after the deadline, to enable others to continue with the best solution of another team

How to pass

- Scope of project ~ 150 hours per student
- Teams of 1-3 students
- Steps can be performed in new group

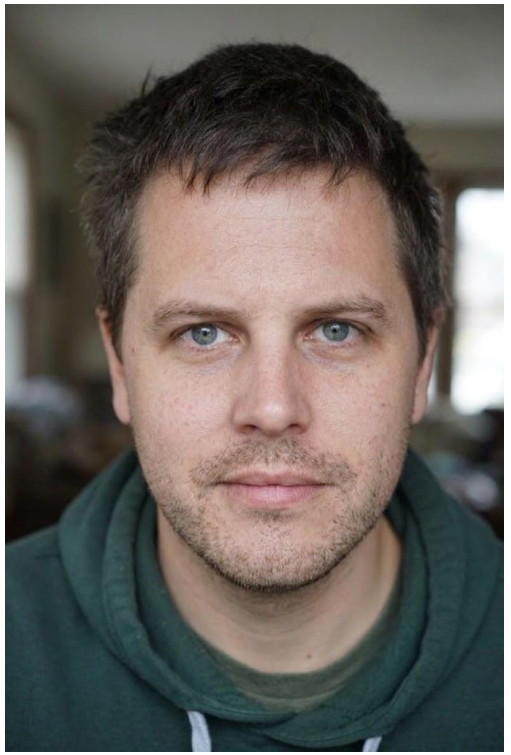
- If you are looking for a group, please stay after the class and talk to people who also stay
- If you are looking for a group and can only join virtually, please use the forum in “Computational Visual Perception” to team up

- Finding a group is your responsibility

Demo

- How to generate Mooney images
- What is a good Mooney image





Part 1 Task 1

- Choose an idea on what type of images/meshes you want to work on
  - keep all of the following tasks in mind for the choice ☺
  - Select/generate 100 images or meshes that are suitable for your choice

- Groups of 3 have to generate images from 3D scenes

Part 1 Task 2

- Write a tool that generates good Mooney image
  - Feel free to use AI, but stay in the driver seat!
- The tool (GUI or CLI) take an image (or a 3D mesh) as input and should have controls for strength of blur and percentage of white pixels
- The tool needs to be able to batch-process 100+ images
- Generate 100 good Mooney images
- Select manually the top 10 out of those 100

Part 1 Task 3

- Choose two computer vision systems (e.g. Monocular Depth estimation and a VLM for Inverse Rendering) and install them on your available machine (choose based on available hardware)
- Run all images through the computer vision systems
  - In their original form
  - As Mooney images

- Groups of 2 or 3 have to present results for four vision systems

Part 1 deliverables

- Per project team 1 single zip file with the following folder structure
  - Original: contains 100 original images
  - Mooney: contains 100 Mooney images
  - Subset:
    - Original: contains selected 10 original images
    - Mooney: contains selected 10 Mooney images
  - Result1:
    - Original: contains results for all original images processed with first vision system
    - Mooney: contains results for all Mooney images processed with first vision system
  - Result2(-4):
    - Original: contains results for all original images processed with nth vision system
    - Mooney: contains results for all Mooney images processed with nth vision system
  - A single pptx file (or powerpoint compatible) containing:
    - Title slide with all team members names
    - 1 page description of your approach/ideas
    - Slide with 10 original images and 10 Mooney images
    - Declaration of used vision system
    - Slide(s) with processing results for each of the vision systems
    - Screenshot showing your tool
    - Screenshot of piece of code that generates Mooneys
    - Declaration of AI usage

Part 1 Upload

- In Studon course :
  “Computational Visual Perception”
- You will upload up to ~1GB (don’t!)
  Plan in internet speed, upload at university

  If you run into issues uploading, you send an md5 hash of your zip file **before** the deadline and you provide an alternative download link within 24h

Part 1 grading

- Correct folder structure and files
- Powerpoint in correct structure
- Mooney Generation as described
- Results for Vision Systems for Mooneys
- Results for Vision Systems for Original Images

- Selected solution: all points fulfilled to full satisfaction
- Pass: at most one bullet point from the above missing

- Groups of 2 or 3 have to present results for four vision systems
- Groups of 3 have to generate images from 3D scenes

- Plagiarism will have serious consequences



Schedule

https://docs.google.com/spreadsheets/d/12SK0zq_UmqIedVcpoxwriRZEq8jYaAnKpcyD_baKGtQ/edit?usp=sharing

Project Consulting

- You can ask questions in the forum
  “Computational Visual Perception”

- You come with concrete questions
- I’ll open a thread in the forum, where you can respond till Thursday before each slot, if you want to meet
- I’ll distribute time slots each Friday

- No guarantee for any responses on the day of the deadline

How to complete module

- 7,5 ECTS
  - 5+2,5 ECTS
  - You need both!
  - You can’t get only 5 or only 2,5 ECTS

Don’t start late

